

🎵 Music Sound Understanding Course (6 Lessons)

Lesson 1: Fundamentals of Sound

Objectives:

- Understand what sound is and how it travels.
- Learn about frequency, amplitude, and waveforms.

Topics:

- Sound waves: frequency (pitch) and amplitude (volume)
- Timbre and overtones
- Waveforms (sine, square, triangle, sawtooth)

Activities:

- Listen to pure tones vs complex sounds.
 - Use a tone generator to explore pitch changes.
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Lesson 2: Musical Pitch and Scales

Objectives:

- Understand pitch perception and tuning systems.
- Learn about the Western musical scale.

Topics:

- Equal temperament vs just intonation
- Octaves, semitones, and tuning (A=440 Hz)
- Scales: major, minor, pentatonic

Activities:

- Identify intervals by ear.
 - Play or generate a major scale and try to sing along.
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Lesson 3: Rhythm and Time

Objectives:

- Understand how rhythm and tempo work.
- Learn about meter and time signatures.

Topics:

- Beat, tempo (BPM), meter (4/4, 3/4, etc.)
- Note values (quarter, eighth, sixteenth notes)
- Syncopation and polyrhythms

Activities:

- Tap along with metronomes at different BPMs.
 - Identify strong and weak beats in popular songs.
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Lesson 4: Timbre and Instrumentation**Objectives:**

- Learn what gives instruments their unique sounds.
- Understand how timbre affects musical texture.

Topics:

- Harmonics and spectral content
- Acoustic vs electronic instruments
- Texture: monophonic, polyphonic, homophonic

Activities:

- Compare violin vs synth vs piano on the same note.
 - Identify instruments in a song by ear.
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Lesson 5: Harmony and Chords**Objectives:**

- Understand the basics of harmony and chord building.
- Learn how chords evoke emotion in music.

Topics:

- Triads, major/minor chords, seventh chords
- Chord progressions (e.g., I-IV-V-I)
- Consonance vs dissonance

Activities:

- Play or listen to I-IV-V-I progression in different keys.
 - Practice recognizing major vs minor chords.
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Lesson 6: Sound Design and Mixing Basics

Objectives:

- Learn how sound is shaped in production.
- Understand basic mixing principles.

Topics:

- EQ, compression, reverb, delay
- Panning and stereo field
- Sound layering and balance

Activities:

- Use free DAW (like Audacity or Cakewalk) to edit basic tracks.
- Experiment with EQ on vocals vs drums.